THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 5	$3^2 + 4^2 = 9 + 16 = 25$, hypotenuse $= 5$
2	**(b)** 4	$5^2 - 3^2 = 16$, other leg $= 4$
3	**(c)** 3, 4, 5	$3^2 + 4^2 = 25 = 5^2$
4	**(d)** 12	Hypotenuse $= 5$; perimeter $= 3 + 4 + 5 = 12$
5	**(a)** 5	Diagonal$^2 = 9 + 16 = 25$
6	**(c)** hypotenuse	Side opposite the right angle
7	**(c)** 10	$6^2 + 8^2 = 36 + 64 = 100$, hypotenuse $= 10$
8	**(a)** 8	$10^2 - 6^2 = 64$, other leg $= 8$
9	**(d)** 6, 8, 10	$6^2 + 8^2 = 100 = 10^2$
10	**(a)** 24	Hypotenuse $= 10$; perimeter $= 6 + 8 + 10 = 24$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$h^2 = 3^2 + 4^2 = 9 + 16 = 25$ (1); $h = 5$ cm (1).	1+1
12	$x^2 = 17^2 - 8^2 = 289 - 64 = 225$ (1); $x = 15$ cm (1).	1+1
13	$6^2 + 8^2 = 100 = 10^2$ (1), so it is right-angled (1).	1+1
14	$h^2 = 6^2 + 8^2 = 36 + 64 = 100$ (1); $h = 10$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Height$^2 = 5^2 - 3^2 = 25 - 9 = 16$ (2); height $= 4$ m (1).	2+1
16	Hypotenuse$^2 = 25 + 144 = 169$, hypotenuse $= 13$ cm (2); perimeter $= 5 + 12 + 13 = 30$ cm (1).	2+1
17	Distance$^2 = 6^2 + 8^2 = 100$ (2); distance $= 10$ km (1).	2+1
	OR: Height$^2 = 10^2 - 6^2 = 100 - 36 = 64$ (2); height $= 8$ m (1).	2+1
18	Hypotenuse$^2 = 81 + 144 = 225$, hypotenuse $= 15$ cm (2); perimeter $= 9 + 12 + 15 = 36$ cm (1).	2+1
	OR: Distance$^2 = 5^2 + 12^2 = 169$ (2); distance $= 13$ km (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonal$^2 = 9 + 16 = 25$, diagonal $= 5$ m (3); distance along two sides $= 7$ m, saved $= 2$ m (2).	3+2
	OR: Triplet: $4, 3, 5$ (2); $16 + 9 = 25$ (2); $5^2 = 25$ so verified (1).	2+2+1
20	Diagonal$^2 = 36 + 64 = 100$, diagonal $= 10$ m (3); distance along two sides $= 14$ m, saved $= 4$ m (2).	3+2
	OR: Triplet: $6, 8, 10$ (2); $36 + 64 = 100$ (2); $10^2 = 100$ so verified (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 10	Diagonal$^2 = 36 + 64 = 100$
2	**(c)** 90°	Right angle
3	**(c)** 13	$5^2 + 12^2 = 25 + 144 = 169$, hypotenuse $= 13$
4	**(c)** 12	$13^2 - 5^2 = 144$, other leg $= 12$
5	**(d)** 5, 12, 13	$5^2 + 12^2 = 169 = 13^2$
6	**(b)** 30	Hypotenuse $= 13$; perimeter $= 5 + 12 + 13 = 30$
7	**(b)** 13	Diagonal$^2 = 25 + 144 = 169$
8	**(d)** $a^2 + b^2 = c^2$	Baudhayana–Pythagoras theorem
9	**(c)** 15	$9^2 + 12^2 = 81 + 144 = 225$, hypotenuse $= 15$
10	**(c)** 12	$15^2 - 9^2 = 144$, other leg $= 12$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 = 25^2 - 7^2 = 625 - 49 = 576$ (1); $x = 24$ cm (1).	1+1
12	$4^2 + 5^2 = 41 \neq 6^2 = 36$ (1), so it is not right-angled (1).	1+1
13	$h^2 = 5^2 + 12^2 = 25 + 144 = 169$ (1); $h = 13$ cm (1).	1+1
14	$x^2 = 20^2 - 12^2 = 400 - 144 = 256$ (1); $x = 16$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Height$^2 = 13^2 - 5^2 = 169 - 25 = 144$ (2); height $= 12$ m (1).	2+1
16	Hypotenuse$^2 = 64 + 225 = 289$, hypotenuse $= 17$ cm (2); perimeter $= 8 + 15 + 17 = 40$ cm (1).	2+1
17	Distance$^2 = 9^2 + 12^2 = 225$ (2); distance $= 15$ km (1).	2+1
	OR: Height$^2 = 15^2 - 9^2 = 225 - 81 = 144$ (2); height $= 12$ m (1).	2+1
18	Hypotenuse$^2 = 49 + 576 = 625$, hypotenuse $= 25$ cm (2); perimeter $= 7 + 24 + 25 = 56$ cm (1).	2+1
	OR: Distance$^2 = 8^2 + 15^2 = 289$ (2); distance $= 17$ km (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonal$^2 = 25 + 144 = 169$, diagonal $= 13$ m (3); distance along two sides $= 17$ m, saved $= 4$ m (2).	3+2
	OR: Triplet: 8, 15, 17 (2); $64 + 225 = 289$ (2); $17^2 = 289$ so verified (1).	2+2+1
20	Diagonal$^2 = 81 + 144 = 225$, diagonal $= 15$ m (3); distance along two sides $= 21$ m, saved $= 6$ m (2).	3+2
	OR: Triplet: 10, 24, 26 (2); $100 + 576 = 676$ (2); $26^2 = 676$ so verified (1).	2+2+1

Mistakes to avoid:

- Applying $c^2 = a^2 + b^2$ with the wrong side as hypotenuse; c is always the side opposite the right angle (the longest).
- Forgetting to take the square root at the end (giving 169 instead of 13).
- Adding the squares when finding a leg: leg $= \sqrt{c^2 - a^2}$ (subtract).
- Using the converse with the wrong pairing: always square the **longest** side on one side of the equation.
- Writing $\sqrt{a^2 + b^2} = a + b$; for example $\sqrt{9 + 16} = 5$, not $3 + 4 = 7$.
- Assuming three consecutive numbers such as 2, 3, 4 form a right triangle; always verify $a^2 + b^2 = c^2$ (here $4 + 9 \neq 16$).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 9: **The Baudhayana–Pythagoras Theorem** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 9, 12, 15	$9^2 + 12^2 = 225 = 15^2$
2	**(d)** 36	Hypotenuse $= 15$; perimeter $= 9 + 12 + 15 = 36$
3	**(a)** 15	Diagonal$^2 = 81 + 144 = 225$
4	**(c)** 3, 4, 5	$3^2 + 4^2 = 5^2$
5	**(b)** 17	$8^2 + 15^2 = 64 + 225 = 289$, hypotenuse $= 17$
6	**(b)** 15	$17^2 - 8^2 = 225$, other leg $= 15$
7	**(a)** 8, 15, 17	$8^2 + 15^2 = 289 = 17^2$
8	**(b)** 40	Hypotenuse $= 17$; perimeter $= 8 + 15 + 17 = 40$
9	**(c)** 17	Diagonal$^2 = 64 + 225 = 289$
10	**(d)** right-angled triangles only	Right triangles only

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$9^2 + 12^2 = 225 = 15^2$ (1), so it is right-angled (1).	1+1
12	$h^2 = 9^2 + 12^2 = 81 + 144 = 225$ (1); $h = 15$ cm (1).	1+1
13	$x^2 = 25^2 - 15^2 = 625 - 225 = 400$ (1); $x = 20$ cm (1).	1+1
14	$5^2 + 7^2 = 74 \neq 9^2 = 81$ (1), so it is not right-angled (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Height$^2 = 17^2 - 8^2 = 289 - 64 = 225$ (2); height $= 15$ m (1).	2+1
16	Hypotenuse$^2 = 144 + 256 = 400$, hypotenuse $= 20$ cm (2); perimeter $= 12 + 16 + 20 = 48$ cm (1).	2+1
17	Distance$^2 = 7^2 + 24^2 = 625$ (2); distance $= 25$ km (1).	2+1
	OR: Height$^2 = 25^2 - 7^2 = 625 - 49 = 576$ (2); height $= 24$ m (1).	2+1
18	Hypotenuse$^2 = 225 + 400 = 625$, hypotenuse $= 25$ cm (2); perimeter $= 15 + 20 + 25 = 60$ cm (1).	2+1
	OR: Distance$^2 = 12^2 + 16^2 = 400$ (2); distance $= 20$ km (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Diagonal$^2 = 64 + 225 = 289$, diagonal $= 17$ m (3); distance along two sides $= 23$ m, saved $= 6$ m (2).	3+2
	OR: Triplet: $12, 35, 37$ (2); $144 + 1225 = 1369$ (2); $37^2 = 1369$ so verified (1).	2+2+1
20	Diagonal$^2 = 49 + 576 = 625$, diagonal $= 25$ m (3); distance along two sides $= 31$ m, saved $= 6$ m (2).	3+2
	OR: Triplet: $14, 48, 50$ (2); $196 + 2304 = 2500$ (2); $50^2 = 2500$ so verified (1).	2+2+1

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